Summary in Graph

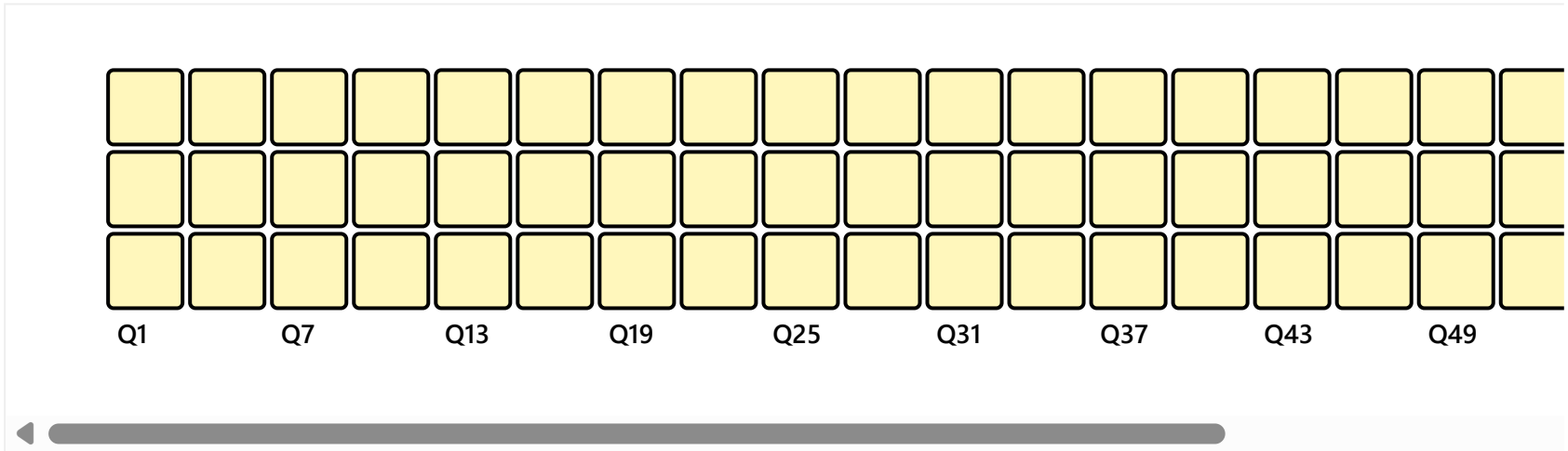
Exam Summary (GO Classes CS 2026 | All India Mock Test 3)

Qs. Attempted:	0 0 + 0	Correct Marks:	0 0 + 0
Correct Attempts:	0 0 + 0	Penalty Marks:	0 0 + 0
Incorrect Attempts:	0 0 + 0	Resultant Marks:	0 0 + 0

Total Questions:	65 30 + 35
Total Marks:	100 30 + 70
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (75) out of 746 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Fill in the blank with the correct form of the verb:

He _____ many of his friends to powerful positions within the government.

A. elevated

B. elevate

C. have been elevating

D. elevating

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Number S is equal to the square of the sum of the digits of a 2 digit number D. If the difference between S and D is 27, then D is

A. 32

B. 54

C. 64

D. 52

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

The average age of A, B and C is 20 years. Two years ago the sum of the ages of A and B was 6 more than that of C. Find the present age of C.

A. 25 years

B. 29 years

C. 28 years

D. 26 years

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

A dinner set is quoted for Rs 1500. A customer pays Rs 1173 for it. If the customer got a series of two discounts and the rate of first discount is 15% then the rate of second discount was:

A. 15%

B. 7%

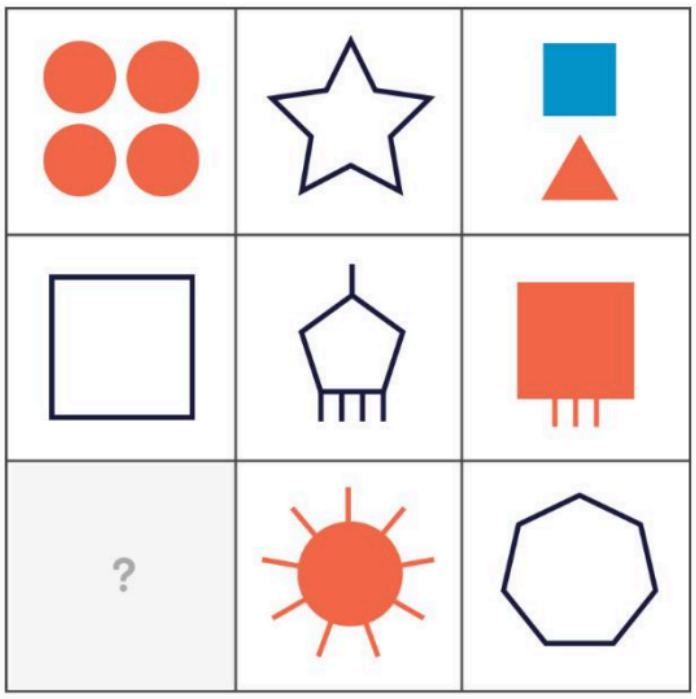
C. 9%

D. 8%

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

The following figure shows a 3×3 grid with one missing entry. Choose the correct option that completes the pattern.



Which of the following options should replace the question mark?

A.

B.

C.

D.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

One percent of the people of country X are taller than 6 ft. Two percent of the people of country Y are taller than 6 ft. There are thrice as many people in country X as in country Y.

Taking both countries together, what is the percentage of people taller than 6 ft?

A. 3.0
B. 2.5
C. 1.5
D. 1.25

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

Let there be a pack of 100 cards numbered 1 to 100. The ith card states:

"There are at most $i-1$ true cards in this pack".

Then how many cards of the pack contain TRUE statements?

A. 0
B. 1
C. 100
D. 50

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

A tank can be filled by two pipes in 24 and 20 minutes, respectively. Another pipe can empty 3 gallons of water per minute from the tank. When all three pipes are working together, it takes 15 minutes to fill the tank. What is the capacity of the tank in gallons?

A. 100

B. 125

C. 130

D. 120

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

In the country of Twenty, there are exactly twenty cities, and there is exactly one direct road between any two cities. No two direct roads have an overlapping road segment. After the election dates are announced, candidates from their respective cities start visiting the other cities.

Following are the rules that the election commission has laid down for the candidates:
Each candidate must visit each of the other cities exactly once.
Each candidate must use only the direct roads between two cities for going from one city to another.
The candidate must return to his own city at the end of the campaign.
No direct road between two cities would be used by more than one candidate.

The maximum possible number of candidates is

A. 9

B. 6

C. 7

D. 8

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

Choose the statement(s) where the underlined word is used correctly:

(i) A prone is a dried plum.

(ii) He was lying prone on the floor.

(iii) People who eat a lot of fat are prone to heart disease.

A. (i) and (iii) only
B. (iii) only
C. (i) and (ii) only
D. (ii) and (iii) only

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 01sec | Discuss

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

An architecture provides an atomic operation `fetch-and-increment(x)`, which atomically returns the current value of `x` and then increments it by 1. The atomic operation is non-preemptible.

The shared variable `inside` is a single shared atomic counter that is initialized exactly once, before any process starts executing the entry protocol. No process reinitializes `inside` upon arrival.

Assume that the value of `inside` never overflows.

```
// shared atomic counter, initialized once before execution
inside = 0;
```

```
// local variable to every process
int a;

do {

  a = fetch-and-increment(inside);

} while (a != 0);

/* critical section */

inside = 0;
```

Which of the following statements is/are TRUE?

A. At most one process can be in the critical section at any time.
B. More than one process may be in the critical section simultaneously.
C. The above solution to the critical section problem is deadlock-free.
D. The above solution to the critical section problem is starvation-free.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

The system consists of 5 processes p_1–p_5 and 4 resource types r_1, r_2, r_3, r_4. Their Allocation, Maximum, and Available resource information is shown below:

Allocation

Process	r_1	r_2	r_3	r_4
p_1	0	0	1	2
p_2	2	0	0	0
p_3	0	0	3	4
p_4	2	3	5	4
p_5	0	3	3	2

Maximum

Process	r_1	r_2	r_3	r_4
p_1	0	0	1	2
p_2	2	7	5	0
p_3	6	6	5	6
p_4	4	3	5	6
p_5	0	6	5	2

Available

$$\begin{bmatrix} r_1 & r_2 & r_3 & r_4 \\ 2 & 1 & 0 & 0 \end{bmatrix}$$

Which of the following is a valid safe sequence for this system?

A. $p_3 \rightarrow p_2 \rightarrow p_5 \rightarrow p_4 \rightarrow p_1$

B. $p_1 \rightarrow p_4 \rightarrow p_5 \rightarrow p_2 \rightarrow p_3$

C. $p_2 \rightarrow p_1 \rightarrow p_3 \rightarrow p_4 \rightarrow p_5$

D. No safe sequence exists.

Your Answer: | **Correct Answer: B** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

Consider a connected network in which each node is connected to exactly three other nodes. The network has a tree-like structure, that is, it contains no cycles. The network uses uncontrolled flooding for broadcast.

Time is divided into discrete steps. In each time step, a node forwards every received broadcast packet to all its neighbors except the one from which it was received. A node does not transmit the same broadcast packet more than once.

At time step 1, a source node sends one broadcast packet to each of its three neighbors. Let $T(t)$ denote the total number of copies of the broadcast packet transmitted at time step t (not up to time step t).

Assume that the network is sufficiently large so that no boundary effects occur within the first 10 time steps. Find the value of $T(10)$.

Your Answer: | **Correct Answer: 1536** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A single packet is sent from **A** to **C** through **B**. Propagation delays are: $A \rightarrow B = d_1$ and $B \rightarrow C = d_2$. Assume $d_1 \leq d_2$, and transmission/queueing delays are negligible.

Flow control **is being implemented in two ways**:

Case 1: End-to-End Flow Control

Flow control operates only between A and C. The packet travels A → B → C, and C sends back an ACK that travels C → B → A. The entire cycle completes only when A receives the ACK from C.

Case 2: Link-by-Link Flow Control

Flow control operates hop-by-hop. When B receives the packet from A, B immediately sends an ACK to A while simultaneously forwarding the packet to C. When C receives the packet, it sends an ACK to B. The entire cycle completes only when A receives B's ACK and B has received C's ACK for the same packet.

What is the total time for **one complete packet cycle** in each case (i.e., time to successfully send one packet from A to C and receive all corresponding ACKs)?

A. End-to-End: $2(d_1 + d_2)$, Link-by-Link: $d_1 + 2d_2$

B. End-to-End: $d_1 + d_2$, Link-by-Link: $d_1 + d_2$

C. End-to-End: $2(d_1 + d_2)$, Link-by-Link: $d_1 + d_2$

D. End-to-End: $2d_2$, Link-by-Link: $2d_1 + d_2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

Assume a TCP sender has an MSS = 100B and a window size of 400B. There is no congestion control so the window size remains constant.

We denote each packet as $p_{\text{seq\#}}$ and the time it is sent as $t_{\text{seq\#}}$. For example, the packet with sequence number 1000 is p_{1000} and is sent at t_{1000}.

Assume that only p_{1400} is lost.

What is the highest ACKed byte after p_{1400} is successfully retransmitted and ACKed, i.e., the highest byte number that the receiver has confirmed as successfully received ?

Your Answer: | Correct Answer: 1799 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Consider the following C program. Assume that the size of an `int` is 4 bytes and the size of a `char` is 1 byte. The program compiles successfully.

```c
#include <stdio.h>
int main(void)
{
    char arr[] = "MachineOrganization";
    int *ptr = (int *)(&arr[1]);
    ptr++;
    printf("value is %c\n", *((char *)ptr));
    return 0;
}
```

What will be the output of the program?

A. value is c

B. value is h

C. value is i

D. value is n

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Programming in C

What will be printed by following C program?

```
#include<stdio.h>
int main() {
    int x = 5, y = 9;
    printf("%d", x ^ y);
    return 0;
}
```

Your Answer: | Correct Answer: 12 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Consider the function definition given below:

```
int f(int a, int b)
{
    if (b == a)
        return 1;
    return a * f(a, b + 1);
}
```

What does $f(2026, 0)$ return?

Assume that your computer can store and manipulate as large integers as necessary without any error.

A. 1
B. 2026
C. 2026^{2025}
D. 2026^{2026}

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Which of the following box-and-arrow diagrams correctly represents the execution of the following **C code**?

```
int x = 5;
int *y = &x;
#define z x
int *w = &z;
```

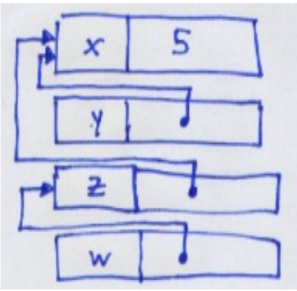
In the diagrams below, each **box** represents a memory location. The left part of a box shows the variable name and the right part shows the value stored in that memory location. For example,

x	5

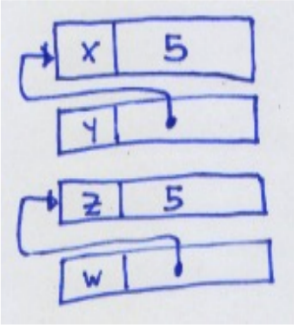
represents that the variable `x` stores the value `5`.

Choose the correct option.

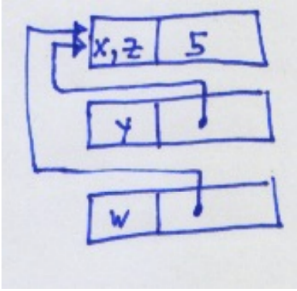
A.



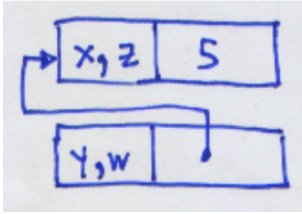
B.



C.



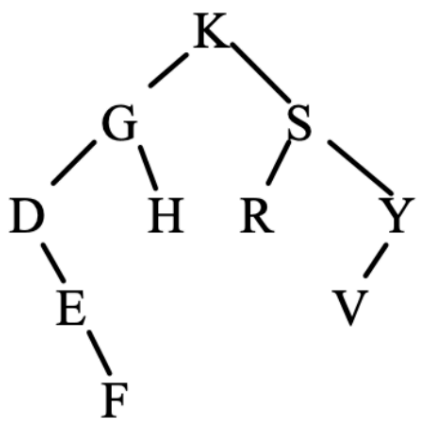
D.



Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following binary tree:



Which of the following is in-order traversal of the tree?

A. E D F G H K R S V Y

B. D E F G H K R S V Y

C. E D F G H R K Y S V

D. D F E G K H R S Y V

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Suppose that only numbers in $1 \ldots 100$ appear as keys in a binary search tree. While searching for 50, which of the following sequences of keys could not be examined?

A. $10, 30, 70, 60, 50$

B. $100, 20, 80, 30, 50$

C. $1, 100, 20, 70, 50$

D. $10, 40, 70, 30, 50$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

A stack is initially empty. We sequentially **push** n elements $(\text{numbered } 1 \text{ to } n)$, where each Push operation takes P seconds, and there is no idle gap between consecutive pushes. After all pushes are completed, the system waits for a random delay W, where W is uniformly distributed in the interval $[0, T]$. Once this waiting period ends, all n elements are **popped** one by one. Each Pop operation also takes P seconds, and there is no gap between consecutive pops.

What is the **expected total time** from the start of the first Push to the moment the stack becomes empty again?

A. $nP + \frac{T}{2} + n$

B. $2nP + \frac{T}{2}$

C. $nP + \frac{T}{2}$

D. $2nP + T$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Let $T(n) = 2T(n/2) + O(n)$, where "O" is big-oh. What will be asymptotic bound on $T(n)$?

A. $\Theta(n)$

B. $\Theta(n \log n)$

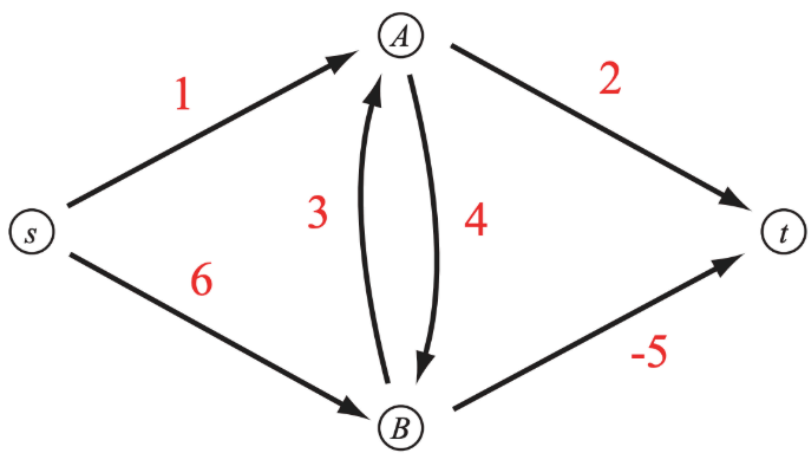
C. $\Theta\left(n^2\right)$

D. $O\left(n^3\right)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Consider the following weighted graph below:



Which of the following options is/are correct about the given graph?

Assume s is the source node for all the options.

A. Dijkstra's algorithm will find the correct shortest path to all the vertices.

B. Dijkstra's algorithm will NOT be able to find the correct shortest path to one or more vertices.

C. Bellman-Ford algorithm will find the correct shortest path to all the vertices.

D. Bellman-Ford algorithm will NOT be able to find the correct shortest path to one or more vertices.

Your Answer: | **Correct Answer: B;C** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider the following system of linear equations:

$$\begin{aligned} x - 3y + 2z &= 5 - a \\ 3x - 4y + z &= a + 1 \\ -3x + 9y - 6z &= 2a \end{aligned}$$

where $a \in \mathbb{R}$. If the given system of linear equations has infinitely many solutions, then what is the value of a ?

Your Answer: | **Correct Answer: 15** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

For a random variable $x(-\infty < x < \infty)$ following a normal distribution, the mean is $\mu = 100$.

If the probability is α for $x \geq 110$, then the probability of x lying between 90 and 110, i.e.,

$$P(90 \leq x \leq 110)$$

will be equal to:

A. $1 - 2\alpha$

B. $1 - \alpha$

C. $1 - \frac{\alpha}{2}$

D. 2α

Your Answer: | **Correct Answer: A** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Select Type | Award: 1 | Penalty: 0 | Digital Logic

Let f be a Boolean function and let $\sim f$ denote the complement of f . In the options given below, "CSOP" denotes Canonical Sum of Products, and "CPOS" denotes Canonical Product of Sums.

Which of the following statements is/are **CORRECT**?

A. The CPOS form of f is the complement of the CSOP form of f .

B. The CSOP form of $\sim \mathrm{f}$ is the complement of the CSOP form of f.

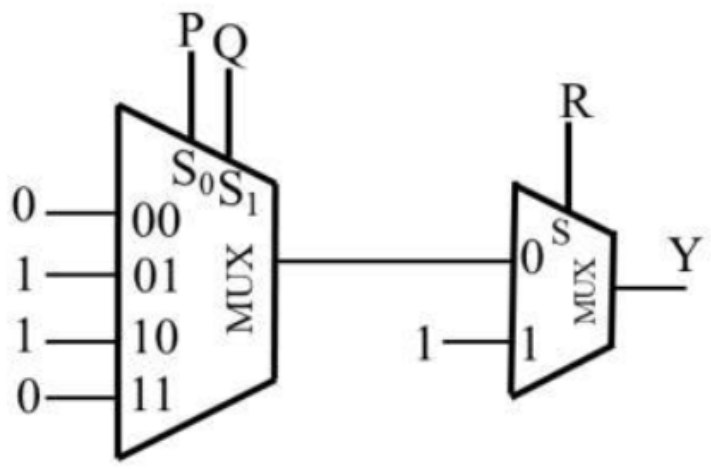
C. The CPOS form of $\sim \mathrm{f}$ is the complement of the CSOP form of f.

D. The CPOS form of $\sim f$ is the CSOP form of f.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

The minimized expression of the Boolean function $\mathrm{Y}(\mathrm{P}, \mathrm{Q}, \mathrm{R})$ implemented by the multiplexer (MUX) circuit shown in the figure is



A. $\mathrm{Y} = \mathrm{R} + (\mathrm{P} \oplus \mathrm{Q})$

B. $\mathrm{Y} = \mathrm{R}(\mathrm{P} \oplus \mathrm{Q})$

C. $\mathrm{Y} = \mathrm{R} + \overline{(\mathrm{P} \oplus \mathrm{Q})}$

D. $\mathrm{Y} = \mathrm{R} \oplus (\mathrm{P} \oplus \mathrm{Q})$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Numerical Type | Award: 1 | Penalty: 0 | CO & Architecture

Suppose we have a processor using an $8-$stage pipeline. If the computer has a $2\ \mathrm{GHz}$ clock, how many nanoseconds does it take to complete each instruction once it enters the pipeline?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Consider a cache with 32-bit addresses, 768 blocks, and a block size of 128 bytes. Tags are 17 bits. How many sets are there, and what is the associativity of the cache?

A. 128 sets, 6-way set associativity.

B. 256 sets, 3-way set associativity.

C. 128 sets, 8-way set associativity.

D. There is not enough information given to answer the question.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

There are two villages X and Y in a faraway land. It is known that each person from village X always tells the truth, and that each person from village Y always lies.

You meet three people A, B, C who are from these villages. You are told that A and B are from the same village. A says, "If B is from village X, then I am from village Y ". Now C says, "If I am from village X, then $2+2=4$".

What can you infer about the villages to which A, B, and C belong?

A. A, B are from village X and C is from village Y.

B. All three of them are from village Y.

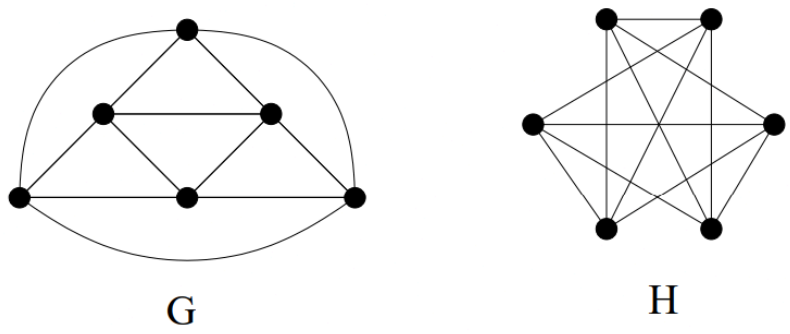
C. A, B are from village Y and C is from village X.

D. The given information is insufficient to infer the villages to which A, B, C belong.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Graph Theory

The graphs $G=(V_G, E_G)$ and $H=(V_H, E_H)$ are **isomorphic.** Indicate the alternative that **justifies this statement.**



A. The degree sequences of the vertices of G and H are the same.

B. The graphs have the same number of vertices and the same number of edges.

C. There exists a **bijection** from V_G to V_H that **preserves adjacency.**

D. Each vertex of G and H belongs to exactly four distinct triangles.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Let $S=\{0,1,2,3,4\}$ be a subset of $\mathbb{Z}$ equipped with the binary operations $\#$ and @.

These operations are defined by the tables given below.

#	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

@	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Based on these operations, consider the following statements:

I. The operation @ satisfies the commutative property.

II. The operation # satisfies the commutative property.

III. In the operation #, 0 is the identity element.

IV. In the operation @, there is no identity element.

Indicate the correct option.

A. Only statements I and II are correct.

B. Only statements II and III are correct.

C. Only statements I, II, and III are correct.

D. All statements are correct.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Which of the following statements is/are true ?

(DFA stands for deterministic finite automaton; NFA stands for nondeterministic finite automaton)

A. For every DFA or NFA, M, over Σ, $L(M) = \Sigma^*$ if and only if each reachable state is an accept state.

B. For all sets A, B, if A and B are both nonregular then $A \cap B$ is also nonregular.

C. For all sets L, L is regular if and only if L^* is regular.

D. For all sets L, L is regular if and only if complement of L is regular.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

In the context of $\mathrm{LALR}(1)$ parsing technique, consider the following statements about the type of items used and the automaton on which the method is based.

Which of the following statements is CORRECT?

A. The $\mathrm{LALR}(1)$ method uses $\mathrm{LR}(0)$ items and is based on the $\mathrm{LR}(0)$ automaton.

B. The $\mathrm{LALR}(1)$ method uses $\mathrm{LR}(1)$ items and is based on the $\mathrm{LR}(0)$ automaton.

C. The $\mathrm{LALR}(1)$ method uses $\mathrm{LR}(0)$ items and is based on the $\mathrm{LR}(1)$ automaton.

D. The $\mathrm{LALR}(1)$ method uses $\mathrm{LR}(1)$ items and is based on the $\mathrm{LR}(1)$ automaton.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following schedules of concurrent read and write operations.

Schedule A

T_a	T_b	T_c	T_d	T_e
	$w(x_2)$			
$w(x_1)$				
			$w(x_3)$	
	$r(x_1)$			
				$r(x_1)$
			$r(x_2)$	
				$w(x_4)$
		$r(x_4)$		
$r(x_3)$				
		$r(x_3)$		
			$r(x_4)$	
	$r(x_4)$			

Schedule B

T_a	T_b	T_c	T_d	T_e
	$w(x_2)$			
$w(x_1)$				
			$r(x_1)$	
	$r(x_1)$			
				$r(x_1)$
			$r(x_2)$	
				$w(x_4)$
		$r(x_4)$		
$r(x_3)$				
		$r(x_3)$		
			$r(x_4)$	
	$r(x_4)$			

Which of these schedules is/are conflict-serializable ?

A. Only schedule A
B. Only schedule B
C. Both
D. None

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Suppose a processor uses a prioritized Round Robin scheduling policy. New processes are assigned an initial quantum of length q. Whenever a process uses its entire quantum without blocking, its new quantum is set to twice its current quantum. If a process blocks before its quantum expires, its new quantum is reset to q. For the purposes of this question, assume that every process requires a finite total amount of CPU time.

Consider the following two schedulers that differ only in how priorities are assigned:

- Scheduler A gives higher priority to processes with larger quanta.
- Scheduler B gives higher priority to processes with smaller quanta.

Which of the following statements about starvation in these schedulers is correct?

A. Scheduler A can cause starvation because large-quantum processes may dominate the CPU.
B. Scheduler B prevents starvation since smaller-quantum processes finish quickly and free the CPU.
C. Scheduler A prevents starvation, but Scheduler B can cause starvation since new processes with quantum q may repeatedly preempt a CPU-bound process whose quantum has doubled.
D. Both schedulers prevent starvation because every process eventually returns to quantum q.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

We use the $P()$ and $V()$ operations on semaphores to prevent race conditions in user-level programs. However, the implementation of $P()$ and $V()$ inside the operating system itself involves incrementing and decrementing the semaphore's count. Since these updates are also shared operations, they could potentially lead to race conditions.

This leads to a natural concern: are we merely relocating the race condition problem to the operating system and reinventing the wheel? How does the operating system *most efficiently* ensure that the internal operations of $P()$ and $V()$ are executed atomically?

Which of the following best describes the correct mechanism?

A. By disabling interrupts in user mode so that $P()$ and $V()$ cannot be interrupted.
B. By protecting only the small internal critical section of $P()$ and $V()$ using hardware-supported atomic instructions.
C. Operating systems execute kernel routines in a controlled environment where concurrent races do not occur.
D. By protecting only the small internal critical section of $P()$ and $V()$ using a purely software-based synchronization mechanism implemented by the operating system, without relying on any special hardware support.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Two cores execute the following threads on shared variables x, y, z.

Initially, $x = 0,\ y = 1,\ z = 2$.

Thread A	Thread B
$x = 10;$	$z = 22;$
$y = 11;$	$y = 21;$
$z = 12;$	$x = 20;$

Each thread follows program order. If a write is overwritten by the other thread, all earlier writes from that thread have no effect. No synchronization is used.

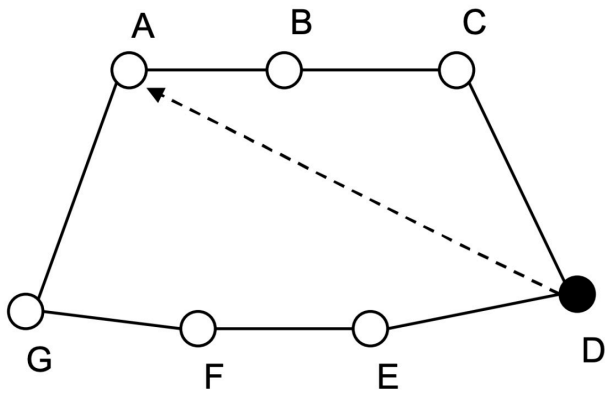
Which final values of (x, y, z) are possible?

A. $(10, 11, 12)$
B. $(20, 11, 12)$
C. $(10, 21, 22)$
D. $(20, 21, 22)$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Consider the network topology shown below. All links in the network are bidirectional and have a cost of 1. The black node D falsely claims that it has a direct link to node A with cost 1, even though such a link does not exist. Routers compute shortest paths using either the distance-vector protocol or the link-state protocol.



In other words, node D is advertising incorrect routing information to the rest of the network. Which one of the following statements is correct?

A. Distance-vector can detect the bogus link, but link-state cannot.
B. Distance-vector cannot detect the bogus link, but link-state can.
C. Both distance-vector and link-state can detect the bogus link.
D. Neither distance-vector nor link-state can detect the bogus link.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Imagine a world where IP addresses are 4 $(\text{not } 32)$ bits long. Suppose a router uses longest-prefix matching and has the following forwarding table:

prefix	output interface
10	A
1	B
x	C
default	D

It is given that a packet with destination address 1001 is forwarded to output interface C. Let:

- P = number of possible values of x for which the above condition holds
- Q = number of distinct 4-bit destination addresses that are forwarded to output interface B

The value of $P+Q$ is _____.

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

Consider the following C program:

```
void swap(int **a, int **b)
{
    int *temp = *a;
    *a = *b;
    *b = temp;
}

int main()
{
    int x = 1, y = 9;

    int *u = &x;
    int *v = &y;

    int **a = &u;
    int **b = &v;

    swap(a, b);

    return 0;
}
```

After the execution of the function call **`swap(a, b)`**, which of the following pairs of variables in `main()` are swapped?

A. x and y
B. u and v
C. a and b
D. Both x and y, and u and v

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider the following C function:

```
Node *zip(Node *a, Node *b) {
    if (a == NULL)
        return b;
    if (b == NULL)
        return a;

    Node *next_a = a->next;
    Node *next_b = b->next;

    a->next = b;
    b->next = zip(next_a, next_b);

    return a;
}
```

Let the two input linked lists be:

$A : 5 \rightarrow 7 \rightarrow 17 \rightarrow \text{NULL}$
$B : 12 \rightarrow 10 \rightarrow 2 \rightarrow 4 \rightarrow 6 \rightarrow \text{NULL}$

Which of the following represents the linked list returned by the function call `zip(A, B)`?

A. $5 \rightarrow 12 \rightarrow 7 \rightarrow 10 \rightarrow 17 \rightarrow 2 \rightarrow 4 \rightarrow 6 \rightarrow \text{NULL}$
B. $5 \rightarrow 7 \rightarrow 17 \rightarrow 12 \rightarrow 10 \rightarrow 2 \rightarrow 4 \rightarrow 6 \rightarrow \text{NULL}$
C. $12 \rightarrow 5 \rightarrow 10 \rightarrow 7 \rightarrow 2 \rightarrow 17 \rightarrow 4 \rightarrow 6 \rightarrow \text{NULL}$
D. $5 \rightarrow 12 \rightarrow 10 \rightarrow 7 \rightarrow 2 \rightarrow 17 \rightarrow 4 \rightarrow 6 \rightarrow \text{NULL}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider a Depth First Search (DFS) traversal and the *start time* and *finish time* assigned to vertices during DFS.

Which of the following statements is/are correct?

A. In a DFS of a directed graph, it is possible to have an edge between two vertices u and v such that u has start and finish times $(10, 40)$ and v has start and finish times $(30, 50)$.

B. In a DFS of an undirected graph, it is possible to have an edge between two vertices u and v such that u has start and finish times $(5, 20)$ and v has start and finish times $(30, 50)$.

C. In a DFS of a directed graph, it is possible to have an edge between two vertices u and v such that u has start and finish times $(5, 20)$ and v has start and finish times $(30, 50)$.

D. In a DFS of a directed graph, if vertex u has start and finish times $(10, 20)$ and vertex v has start and finish times $(15, 18)$, then the edge (v, u) must be a back edge.

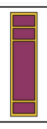
Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider the following unsorted array, and the same array after 4 iterations of insertion sort (where we sweep through the array from left to right, and move an element to its relative correct position).

Assume no two elements are equal. Insertion sort arranges the elements from least to greatest.

Which of the following is necessarily the correct relation <, or > between objects?

	Symbol 1		Symbol 2
A.		<	
B.		>	
C.		<	
D.		>	

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following grammar in which S is the start non-terminal and in which the productions for the non-terminals A, B and C are not shown. There are no other productions for S.

$$S \to AB \mid AC$$
$$A \to \ldots$$
$$B \to \ldots$$
$$C \to \ldots$$

Consider now the $\mathrm{LL}(1)$ parsing table for this grammar. What conditions must be satisfied by the First and Follow sets of the non-terminals, such that the row corresponding to S in this table does not contain multiply-defined entries?

(Select all the necessary conditions. Each option is to be considered independently.)

A. $\mathrm{FIRST}\{a\} = \{\varepsilon\}$
B. $\mathrm{First}(B) \cap \mathrm{First}(C) = \varnothing$
C. $\mathrm{First}(B) \cap \mathrm{Follow}(S) = \varnothing$
D. If $\varepsilon \in \mathrm{First}(B)$ and $\varepsilon \notin \mathrm{First}(C)$, then $\mathrm{First}(C) \cap \mathrm{Follow}(S) = \varnothing$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider the joint probability mass function of random variables X and Y as shown in the table below. For instance, $P(X = 1, Y = 2) = 0.3$.

	$X = 1$	$X = 2$
$Y = 1$	0.2	0.3
$Y = 2$	0.3	0.1
$Y = 3$	0.1	

The value of $P(X = 2 \mid Y = 2)$ is:

A. 0.10
B. 0.25
C. 0.40
D. 0.75

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

Consider the following boolean function $f(A, B, C, D)$:

$$f \;=\; A'BC \;+\; (AC + B)D \;+\; h(A, B, C, D)$$

The boolean function $h(A, B, C, D)$ is unknown. The total number of Boolean functions h, that will make the function f self-dual, is ?

Your Answer: | Correct Answer: 256 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

You have to make a $2^n : 1$ MUX (A multiplexer with 2^n inputs and 1 output) using only $2 : 1$ MUXes. The number of $2 : 1$ MUXes needed is (only one answer is correct):

A. $n - 1$, $2 : 1$ MUXes will be needed.
B. 2^{n-1}, $2 : 1$ MUXes will be needed.
C. 2^n, $2 : 1$ MUXes will be needed.
D. $2^n - 1$, $2 : 1$ MUXes will be needed.

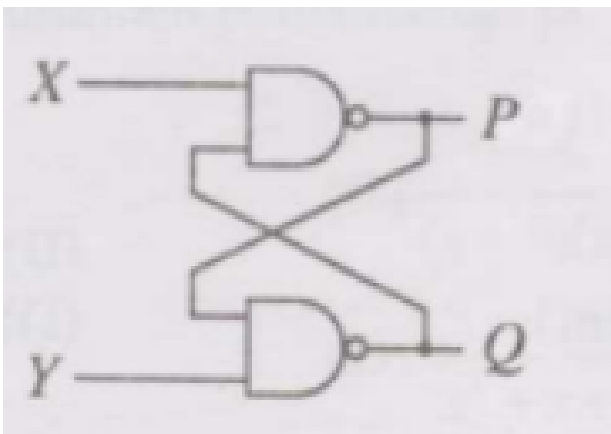
Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The following binary values were applied to the X and Y inputs of the NAND latch shown in the figure in the sequence indicated below:

$$\mathrm{X} = 0, \mathrm{Y} = 1; \quad \mathrm{X} = 0, \mathrm{Y} = 0; \quad \mathrm{X} = 1, \mathrm{Y} = 1.$$

The corresponding stable P, Q outputs will be



A. $\mathrm{P} = 1, \mathrm{Q} = 0;$ $\mathrm{P} = 1, \mathrm{Q} = 0;$ $\mathrm{P} = 1, \mathrm{Q} = 0$ or $\mathrm{P} = 0, \mathrm{Q} = 1$

B. $\mathrm{P} = 1, \mathrm{Q} = 0;$ $\mathrm{P} = 0, \mathrm{Q} = 1$ or $\mathrm{P} = 0, \mathrm{Q} = 1;$ $\mathrm{P} = 0, \mathrm{Q} = 1$

C. $\mathrm{P} = 1, \mathrm{Q} = 0;$ $\mathrm{P} = 1, \mathrm{Q} = 1;$ $\mathrm{P} = 1, \mathrm{Q} = 0$ or $\mathrm{P} = 0, \mathrm{Q} = 1$

D. $\mathrm{P} = 1, \mathrm{Q} = 0;$ $\mathrm{P} = 1, \mathrm{Q} = 1;$ $\mathrm{P} = 1, \mathrm{Q} = 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Suppose we have an 8-stage pipeline that can complete one instruction each clock cycle in the absence of any jumps or branches. (We assume there are no data or structural hazards.) For jumps or branches, the destination of the jump is known at the end of the pipeline's second stage, whereas the decision of whether a branch is taken is determined at the end of the pipeline's fourth stage.

Assume that 5% of instructions in a typical program are jump instructions (i.e. unconditional branch instructions), while 15% of instructions are branch instructions(i.e. conditional branch instructions), of which $2/3$ are taken.

Let M be the average number of clock cycles per instruction if we speculatively execute assuming that branches are not taken.

Let N be the average number of clock cycles per instruction if we speculatively execute assuming that branches are taken. What are M, N, respectively ?

A. $1.5, 1.3$

B. $1.3, 1.35$

C. $1.35, 1.3$

D. $1.35, 1.5$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a 4096-byte direct-mapped cache with 16 byte blocks and 256 blocks. Initially, all of the blocks are invalid. The physical address size is 32 bits.

Suppose the following sequence of memory addresses is accessed:

```
    0x0567
    0x03C6
    0x18B7
    0x0873
 5. 0x087C
    0x0872
    0x18EC
    0x1874
    0x18BA
10. 0x17AB
    0x0870
    0x056C
    0x18E1
    0x0146
15. 0x02C5
    0x02C2
```

How many of these memory accesses will result in a cache miss?

Your Answer: | Correct Answer: 10 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A computer scientist needs to evaluate the performance of his computer in order to calculate the response time of an algorithm. Considering that all the sectors of track 0 are read starting from sector 0, then all the sectors of track 1 starting from sector 0, and so on. The disk rotates once in 30 ms, and a seek takes 20 ms between adjacent cylinders and 80 ms in the worst case. Switching between tracks within the same cylinder can be done instantaneously. Assume that initially the disk head is at an arbitrary cylinder, and ignore the rotational delay i.e. the time it takes for the beginning of the sector 0 to reach under the disk head (After every seek or track switch, the disk is assumed to be positioned at sector 0).

In this way, how much time is required to read a disk with $2,400$ cylinders, each with six tracks of 32 sectors, in seconds?

A. 628.12

B. 480.06

C. 320.12

D. 268.06

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Numerical Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

A relation R on the set $A = \{a, b, c, d\}$ is defined by reading the columns of the following table from top to bottom. If a column in the table reads $(x, y, 1)$ it means x is related to y in R. If a column in the table reads $(x, y, 0)$ it means x is not related to y.

a	b	c	d	a	b	c	d	a	b	c	d	a	b	c	d
a	a	a	a	b	b	b	b	c	c	c	c	d	d	d	d
0	0	1	0	1	0	0	0	1	1	0	1	0	0	1	0

For instance, from the fifth column we have, $(a, b) \in R$, and from the second we have $(b, a) \notin R$.

Another relation S on the set A is defined as: for any $x, y \in A$, the pair (x, y) is in S if and only if there exists $z \in A$ such that *both* $(x, z) \in R$ and $(z, y) \in R$ hold.

The cardinality of S is?

Your Answer: | Correct Answer: 9 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

A graph consists of a finite set of vertices, and edges between some (unordered) pairs of these vertices. The graphs in this question have no loops (an edge between a vertex and itself) or multiple edges (more than one edge between the same pair of vertices). A vertex v in a graph is said to be a global vertex if there is an edge between v and every other vertex in the graph. A graph G is constructed as follows: First, its vertex set is assigned to be the set $\{v_1, v_2, \ldots, v_{10}\}$. Next, between each pair of distinct vertices v_i, v_j, an edge is added with probability $\frac{1}{2}$. Let the expected number of global vertices in graph G be N. What is $\mathrm{N} * 2^{10}$?

Your Answer: | Correct Answer: 20 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following problems is/are decidable?

A. Given a context-free language L and a regular language R, is $L \subseteq R$?

B. Given a context-free language L and a regular language R, is $R \subseteq L$?

C. Given a DFA, does it accept on only finitely many inputs?

D. Given a PDA, does it accept on only finitely many inputs?

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Consider the following context-free grammar G over the alphabet $\Sigma = \{a, b\}$.

$$S \to aB \mid Ab \mid ab$$
$$A \to aS$$
$$B \to Sb$$

Let $\mathrm{G_A}$ denote the grammar obtained from G by deleting the variable A and all its related productions. Similarly, $\mathrm{G_B}$ is defined.

Which of the following is/are true?

A. G is ambiguous.

B. $\mathrm{G_A}$ is ambiguous.

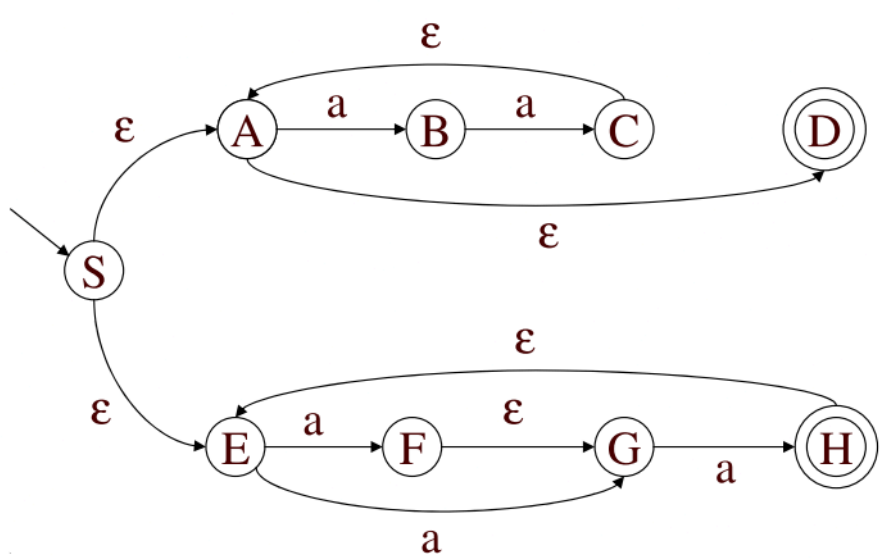
C. $\mathrm{L(G) = L\left(G_A\right)}$

D. $\mathrm{L(G) = L\left(G_B\right)}$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Numerical Type | Award: 2 | Penalty: 0 | Theory of Computation

Consider the following NFA over the alphabet $\{\mathrm{a}\}$.

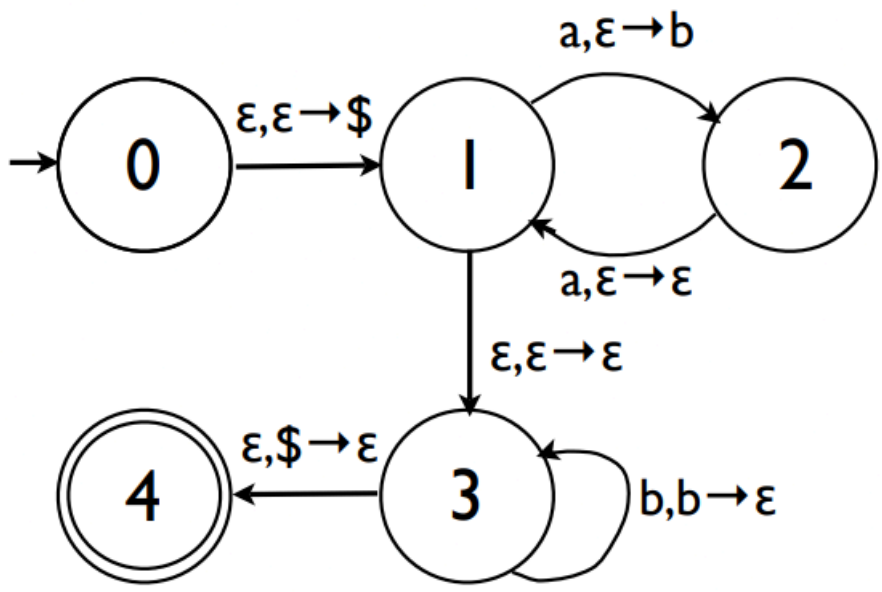


The minimum number of states in an equivalent DFA is?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Numerical Type | Award: 2 | Penalty: 0 | Theory of Computation

Let M be the following PDA(push-down automaton):



The number of strings of length 100 accepted by the above pushdown automaton is?

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Regarding timestamp-ordering-based concurrency control, let

- $r_ts(x)$ denote the largest timestamp of any transaction that has successfully read the data item x, and
- $w_ts(x)$ denote the largest timestamp of any transaction that has successfully written the data item x.

Consider a transaction T with timestamp $\text{ts}(T)$ that needs to write the data item x, that is, T issues the operation $\text{write}(x)$. According to the basic timestamp-ordering algorithm, in order for the schedule to be conflict-serializable without aborting and restarting T, the following predicate must be true:

A. $(r_ts(x) > \text{ts(T)})$ or $(w_ts(x) > \text{ts(T)})$

B. $(r_ts(x) > \text{ts(T)})$ and $(w_ts(x) > \text{ts(T)})$

C. $(\text{not}(r_ts(x) > \text{ts(T)}))$ or $(\text{not}(w_ts(x) > \text{ts(T)}))$

D. $(\text{not}(r_ts(x) > \text{ts(T)}))$ and $\text{not}((w_ts(x) > \text{ts(T)}))$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Let R be the set of attributes of relation instance r and S the set of attributes of relation instance s such that $\text{S} \subset \text{R}$.

In the relational algebra, consider an operation "$\#$", defined as below: $\text{r}\#\text{s}$ is defined to be the largest relation q such that $\text{q} \times \text{s} \subseteq \text{r}$.

Instances r, s of the schema R, S, respectively, are given below.

r			
A	B	X	Y
a	b	w	x
c	d	y	z
a	b	y	z
e	f	w	x
c	d	w	x

s	
X	Y
y	z
w	x

The number of rows in $\mathrm{r}\#\mathrm{s}$?

Your Answer: | **Correct Answer: 2** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider a relation $\mathbf{R}$ with $3d$ attributes (d is an integer ≥ 2) that are named

$$A_0, A_1, A_2, \ldots, A_{3d-1}$$

There is a set $\mathbf{F}$ of $3d$ functional dependencies on $\mathbf{R}$ of the form:

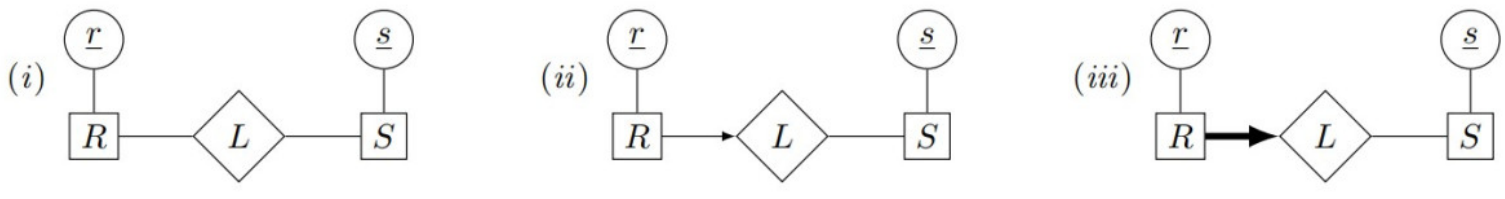
$$A_i \rightarrow A_{(i+3) \bmod 3d}, \quad \text{for } i = 0, 1, 2, \ldots, 3d-1$$

Here, mod denotes the modulo operator (remainder after integer division). For $\mathrm{d} = 7$, What is the number of keys in R?

Your Answer: | **Correct Answer: 343** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following ER diagrams labelled i, ii, and iii.



Consider the following SQL statements.

(p)

```
CREATE TABLE S(s INT, PRIMARY KEY (s));
CREATE TABLE R(r INT, s INT REFERENCES S, PRIMARY KEY (r));
```

(q)

```
CREATE TABLE R(r INT, PRIMARY KEY (r));
CREATE TABLE S(s INT, PRIMARY KEY (s));
CREATE TABLE L(r INT REFERENCES R, s INT REFERENCES S);
```

(r)

```
CREATE TABLE S(s INT, PRIMARY KEY (s));
CREATE TABLE R(r INT, s INT NOT NULL REFERENCES S, PRIMARY KEY (r));
```

Which ER diagram corresponds to which SQL statement?

A. (i) - (p), (ii) - (q), (iii) - (r)
B. (i) - (q), (ii) - (p), (iii) - (r)
C. (i) - (q), (ii) - (r), (iii) - (p)
D. (i) - (r), (ii) - (p), (iii) - (q)

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the following program. Assuming that all operations take their operands from registers, Minimum how many registers would be needed to perform register allocation with no spilling?

1. $A = 7;$
2. $B = A + 2;$
3. $C = A + B;$
4. $D = C + B;$
5. $B = C + B;$
6. $A = A + B;$
7. $E = C + D;$
8. $F = C + D;$
9. $G = A + B;$
10. $H = E + F;$

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Which of the following statements regarding SSA form (Static Single Assignment form) of an intermediate representation is False?

A. In the SSA-form for a procedure, it is LEGAL for a compiler to reorder two consecutive assignment statements $\mathrm{S1}$ and $\mathrm{S2}$ within the same basic block if $\mathrm{S2}$ does NOT use the value computed by $\mathrm{S1}$ (and the statements have no other side-effects).

B. In the Non-SSA form for a procedure, it is LEGAL for a compiler to reorder two consecutive assignment statements $\mathrm{S1}$ and $\mathrm{S2}$ within the same basic block if $\mathrm{S2}$ does NOT use the value computed by $\mathrm{S1}$ (and the statements have no other side-effects).

C. Converting a program into SSA eliminates false dependencies (anti and output) from the program, but preserves true dependences.

D. For a procedure in SSA form, every SSA variable is written exactly once in each invocation of the procedure.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

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